APM1110: Probability and Probability Distributions

Formative Assessment 8 – Solutions

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Problem 1

The urn has 10 red, 8 black and 7 green balls (25 in total). Each color is chosen with probability $\frac{1}{3}$, then 4 balls are drawn without replacement.

(a) Given a chosen color, $X = 0$ means all 4 balls come from the other colors, which number 15, 17 and 18 for red, black and green.

$$P(X=0) = \frac{1}{3} \cdot \frac{\binom{15}{4} + \binom{17}{4} + \binom{18}{4}}{\binom{25}{4}} = \frac{1}{3} \cdot \frac{1365 + 2380 + 3060}{12650} = \boxed{\frac{1361}{7590} \approx 0.1793}$$

(b) Given the chosen color, the ith ball drawn is equally likely to be any of the 25 balls, so

$$P(X_i = 1) = \frac{1}{3}\left(\frac{10}{25} + \frac{8}{25} + \frac{7}{25}\right) = \boxed{\frac{1}{3} \approx 0.3333}, \qquad i = 1, 2, 3, 4.$$

(c) $X = X_1 + X_2 + X_3 + X_4$, so by linearity of expectation

$$E[X] = \sum_{i=1}^{4} P(X_i = 1) = 4 \cdot \frac{1}{3} = \boxed{\frac{4}{3} \approx 1.3333}.$$

Problem 2

$P(X = n) = \dfrac{4}{n(n+1)(n+2)}$, $n \geq 1$.

(a) By the hint,

$$\frac{4}{n(n+1)(n+2)} = 2\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right].$$

Using $\frac{1}{m(m+1)} = \frac{1}{m} - \frac{1}{m+1}$ (the hint with $k = 1$), the series telescopes:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right) = 1, \qquad \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} = \sum_{n=1}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{1}{2}.$$

Therefore

$$\sum_{n=1}^{\infty} P(X=n) = 2\left[1 - \frac{1}{2}\right] = \boxed{1}.$$

Every term is positive, so this is a valid pmf. ■

(b) The factor n cancels:

$$E[X] = \sum_{n=1}^{\infty} n \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} = 4 \cdot \frac{1}{2} = \boxed{2}$$ ■

(c)

$$E[X^2] = \sum_{n=1}^{\infty} n^2 \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)}.$$

For $n \geq 1$, $n+1 \leq 2n$ and $n+2 \leq 3n$, so $\dfrac{n}{(n+1)(n+2)} \geq \dfrac{1}{6n}$. Then

$$E[X^2] \geq \frac{4}{6}\sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

since the harmonic series diverges. Hence

$$\boxed{E[X^2] = \infty}$$ ■

Problem 3

$f(x) = a + bx^2$ on $[0,1]$, with $E[X] = \frac{3}{5}$.

Total probability must be 1, and the mean must be $\frac{3}{5}$:

$$\int_0^1 (a + bx^2)\,dx = a + \frac{b}{3} = 1, \qquad \int_0^1 x(a + bx^2)\,dx = \frac{a}{2} + \frac{b}{4} = \frac{3}{5}.$$

From the first equation, $a = 1 - \frac{b}{3}$. Substituting into the second:

$$\frac{1}{2} - \frac{b}{6} + \frac{b}{4} = \frac{3}{5} \;\Rightarrow\; \frac{b}{12} = \frac{1}{10} \;\Rightarrow\; b = \frac{6}{5}, \qquad a = 1 - \frac{2}{5} = \frac{3}{5}.$$

$$\boxed{a = \tfrac{3}{5} = 0.6000, \qquad b = \tfrac{6}{5} = 1.2000}$$

Problem 4

Write $Y = \dfrac{X - \mu}{\sigma} = \dfrac{1}{\sigma}X - \dfrac{\mu}{\sigma}$.

$$E[Y] = \frac{E[X] - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0, \qquad \mathrm{Var}(Y) = \frac{1}{\sigma^2}\mathrm{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1.$$

$$\boxed{E[Y] = 0, \qquad \mathrm{Var}(Y) = 1}$$

Problem 5

$X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(3)$ are independent, and $W = X + Y$.

(a) A Poisson(λ) variable has MGF $M(t) = e^{\lambda(e^t-1)}$. By independence,

$$M_W(t) = M_X(t)\, M_Y(t) = e^{2(e^t-1)}\, e^{3(e^t-1)} = \boxed{e^{5(e^t-1)}}.$$

(b) This is the MGF of a Poisson distribution with parameter 5. By the uniqueness property of MGFs,

$$\boxed{W \sim \text{Poisson}(\lambda = 5)}.$$

(c) Differentiating $M_W(t) = e^{5(e^t-1)}$:

$$M_W'(t) = 5e^t\, e^{5(e^t-1)} \;\Rightarrow\; E[W] = M_W'(0) = 5,$$

$$M_W''(t) = 5e^t\, e^{5(e^t-1)} + 25e^{2t}\, e^{5(e^t-1)} \;\Rightarrow\; E[W^2] = M_W''(0) = 5 + 25 = 30.$$

$$\boxed{E[W] = 5.0000, \qquad E[W^2] = 30.0000}$$